SemanticSplat: Graph-Pruned Semantic Search

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Abstract. Built-environment digital twins based on 3D Gaussian Splatting (3DGS) are photorealistic but lack an explicit semantic layer for natural-language querying. We present SemanticSplat, a graph-pruned semantic search layer that organizes captured views into a hierarchical zone–region–object tree and answers queries via top-down traversal instead of exhaustive flat search over the same semantic items. On five manually captured indoor/outdoor digital-twin pilot scenes (97 views; 150 queries), graph traversal cuts average views checked from 19.4 to 4.77 (75.5%) and estimated input-context tokens from 3168 to 1014 (68.2%). Under stub lexical matching with verified view labels (n=125), hit@1 is 0.68 (graph) vs. 0.768 (flat), exposing a quality–cost tradeoff. On a 48-query ScanNet oracle-map pilot, graph search matches flat lexical Recall@1 / Acc@0.25 (0.271) while checking about 77% fewer objects. We additionally execute native ConceptGraphs on all eight ScanNet and all eight Replica scenes, plus a reduced-resource end-to-end LangSplat pilot on one ScanNet scene. These external runs use different map-construction protocols and are diagnostic references, not evidence of method superiority.

Keywords: Digital twins · 3D Gaussian Splatting · Scene graphs · Semantic search · 3D object grounding.

1 Introduction

Digital twins of the built environment increasingly rely on neural rendering, including 3D Gaussian Splatting [7], for interactive visualization. Applications in the built environment include facility queries, robotics hand-off, and semantic mapping. They need more than geometry and appearance: a user should ask "where can I sit near the reception?" and receive a grounded viewpoint or object. Exhaustive flat search over all views or objects is simple but wastes context and scales poorly with scene size.

Language-field methods [8,11] distill features into the field but do not expose hierarchical pruning. Object-centric scene graphs such as ConceptGraphs [2] and BBQ [9] provide strong public-dataset object grounding; BBQ in particular reports Acc@k on Replica/ScanNet and Sr3D+/Nr3D/ScanRefer. Our focus is complementary: a hierarchical semantic index over 3DGS-captured views with measured *query-time* context pruning, evaluated under a same-input graph-vs-flat protocol and aligned with BBQ-style Acc@k / Recall@1 on public-dataset pilots.

Contributions.

- A representation-agnostic semantic-map query layer for 3DGS-backed digital twins with hierarchical traversal and affordance expansion.
- Same-input graph-vs-flat evidence on five captured scenes and BBQ-aligned object-grounding pilots on official Replica (8 scenes) and ScanNet (8 scenes).
- External feasibility references: native ConceptGraphs on eight ScanNet and eight Replica scenes and reduced-resource LangSplat on one ScanNet scene; BBQ remains a metric guide because its official code is not run.

2 Related Work

Language fields and 3DGS semantics. LERF [8] embeds language in a radiance field, while LangSplat [11], Semantic Gaussians [3], and LEGS [13] attach open-vocabulary features to Gaussian representations. These methods optimize a different native representation than tree traversal over shared semantic items.

Object-centric and hierarchical scene graphs. ConceptGraphs [2], BBQ [9], HOV-SG [12], and Hydra [4] construct graphs for grounding, navigation, or robotics. BBQ is the closest public-dataset grounding reference: object IDs, captions, 3D extents, spatial relations, and deductive LLM reasoning over a compact subgraph [9]. We cite BBQ Acc@k tables as literature context, not as measurements from this repository.

Open-vocabulary maps and grounding evaluation. ConceptFusion [5] and OpenScene [10] provide open-set or open-vocabulary 3D representations. They do not explicitly prune a zone–region hierarchy. Language grounding in RGB-D scans was established by ScanRefer [1], while OpenLex3D [6] emphasizes tiered evaluation of open-vocabulary 3D representations. Our two evaluation tracks therefore separate view-label retrieval from object-ID and 3D-box grounding.

Positioning. SemanticSplat targets hierarchical context control for 3DGS digital twins. Measured comparisons are same-input graph vs. flat lexical/embedding on identical scenes, items, queries, and labels. We do not claim superiority over BBQ without a same-protocol official-code run.

3 Method

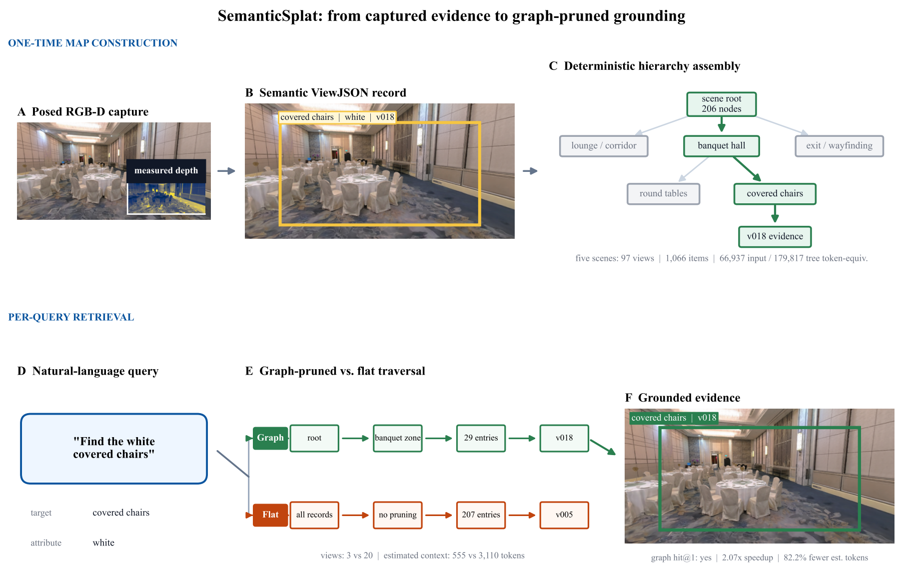

Fig. 1: SemanticSplat end to end, rendered from project evidence rather than illustrative assets. Top: the captured Conference Hall v018 RGB-D observation, its manual ViewJSON record and coarse depth-projected box, and a slice of the checked-in hierarchy. Bottom: saved query qv2_010 is evaluated with the same scorer under graph pruning and flat search, then linked to its returned view. Context-token values are deterministic text-size estimates, not API billing.

3.1 Hierarchical query representation

SemanticSplat starts at query time from a pre-built semantic tree $H = (V, E)$ and its associated semantic records; input construction is described in Sec. 4.1. Nodes carry a type, name, summary, attributes, view IDs, and parent/child links. The rooted hierarchy may contain zones, regions, objects, landmarks, signs, and facilities. Each searchable leaf is linked to its ancestors, so a branch decision defines a candidate subset without altering the leaf records or scorer. Graph and flat methods always receive the same records; they differ only in which records are inspected.

Every node v has type $\tau(v)$. Its ordered path from the scene root is $A(v) = (v_0, \ldots, v_d)$. Searchable types are partitioned into branch nodes (zones and regions), entity nodes (objects, landmarks, signs, and facilities), and evidence nodes (view observations). Parent–child edges encode containment or evidence attachment; spatial relations remain typed attributes rather than being conflated with hierarchy. This separation lets the branch gate reduce the candidate universe while the same entity scorer remains valid at every retained leaf.

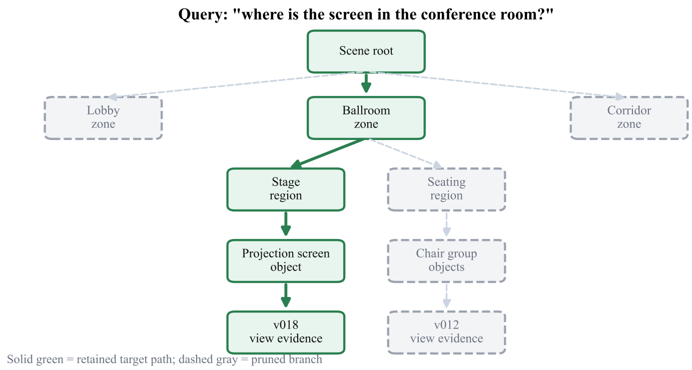

Fig. 2: Hierarchical semantic tree used at query time. A scene root expands through zone, region, object, and view-observation levels. The example query retains the solid green target branch and prunes the dashed gray branches.

For object grounding, the canonical candidate is

$$c_i = (\mathrm{id}_i, \tau_i, A_i, y_i, x_i, R_i, B_i, V_i), \tag{1}$$

where y_i is the label, x_i searchable text, R_i stored relations, $B_i = (B_i^{2D}, B_i^{3D})$ the optional boxes, and V_i supporting views. Source instance IDs are retained when available; otherwise an ID is derived deterministically from the tree node or view observation. Tree-only object nodes remain searchable for backward-compatible indexes. Canonicalization is performed once per scene and its construction cost is excluded from per-query latency.

3.2 Captured-scene zone pruning

The internal five-scene benchmark operates over tree nodes. Query text is lower-cased, stop words are removed, simple aliases are normalized, and plural variants are retained. For each zone b, searchable text concatenates the zone record and the records of its assigned children. With query-token set Q and zone-token set T_b, the gate score is

$$u_b = |Q \cap T_b|. \tag{2}$$

If $u_m = \max_b u_b > 0$, graph search keeps zones satisfying $u_b \geq \max(1, 0.5u_m)$; if no zone overlaps, it keeps every zone. Unzoned entries are always retained. Leaves are ranked by overlap count, normalized overlap, node-type priority (object > landmark > region), and a stable ID tie-break. If no retained leaf overlaps, graph search scans the remaining entries once. Flat search ranks every non-root/non-zone entry with the identical rule. This is the exact implementation behind the internal view/context results, rather than a learned retriever.

3.3 Object-level lexical and intent scoring

The public-dataset grounding track uses a richer object scorer. Let q_t be the parsed target phrase, Q its base tokens, and Q_e the union of base tokens, a fixed synonym map, and deterministic intent expansions. For example, *sit* activates sofa, chair, bench, couch, armchair, lounge, and seating; *exit*, *charge*, *help*, and *restroom* activate their corresponding facility terms. For each top-level branch, $h_b = |Q_e \cap T_b| / \max(1, |Q_e|)$. The graph keeps branches with $h_b \geq \rho \max_j h_j$ using $\rho = 0.5$; with no hierarchy evidence, graph-only search deterministically keeps one branch so fallback behavior remains measurable. Candidate i then receives

$$s_i = \min(1, 0.65e_i + 0.55\ell_i + 0.20t_i + 0.25a_i)\,, \tag{3}$$

where e_i is exact-label match and ℓ_i, t_i, a_i are normalized label, record-text, and synonym/affordance overlaps. Every component and fired affordance key is saved. Flat lexical applies the same scorer to the full candidate universe.

3.4 Target–anchor spatial reasoning

The parser recognizes *near*, left/right, above/below, and front/behind in infix and prefix forms, yielding (q_t, r, q_a). Anchors are retrieved from the full candidate universe even when the target was branch-pruned. A relation can be established by an explicit stored relation string or by geometry. We prefer a valid 3D box and otherwise use a normalized 2D box; boxes with missing or non-finite coordinates, non-positive extent, or coordinates outside declared bounds are rejected.

For *near*, center distance normalized by the mean combined box extent must be at most 1.5. Directional predicates compare the relevant center axis with a 5% combined-extent tolerance; front/behind is unresolved for 2D-only geometry. Let $r_i = 1$ for a satisfied relation, -0.35 for a resolved violation, and 0 when unresolved. Reranking applies

$$s_i \leftarrow \min(1, \max(0, s_i + 0.35r_i)). \tag{4}$$

The result records the anchor, relation score, evidence source, geometric margin, and whether box quality prevented a decision.

3.5 Fallback, variants, and measured cost

Graph+fallback expands to the full candidate universe when the best score is below 0.45, the relation is unresolved, or required box geometry is rejected. The expansion is explicit rather than a silent mode change. We evaluate graph, graph+fallback, flat lexical, and flat embedding, plus orthogonal no-hierarchy, no-relation, and no-fallback ablations. Flat embedding uses the local CPU BGE-small-en-v1.5 encoder and cosine similarity; an unavailable encoder produces a blocked result rather than a lexical substitution.

Each query emits selected object/node/view IDs, ancestor traversal, checked objects/views/nodes, fallback reasons, runtime, serialized-context size, token estimate or measured local tokenizer count, and a 3D box when the method natively provides one. Map/index construction time is recorded separately from per-query search. If N is the full candidate count and N_q the retained count, flat scoring is $O(N)$ and graph leaf scoring is $O(N_q)$ after a gate over the top-level nodes; fallback has the flat worst case.

4 Datasets and Evaluation

4.1 Input representation and index construction

Posed RGB(-D) observations or existing dataset annotations are converted into structured records containing objects, landmarks, room cues, relations, and short summaries. The five captured scenes use manual ViewJSON records and manual zone references; their semantic index is not independent GT. Replica and ScanNet use official object IDs and 3D boxes to construct oracle semantic candidates for the SemanticSplat grounding pilots. These frozen records form the shared input for graph and flat variants (Fig. 1).

Across the five captured scenes, deterministic graph construction rebuilds 97 ViewJSON records (1,066 semantic items) into 1,064 nodes with **zero provider calls and zero provider input/output tokens**. On the evaluation workstation, the sum of per-scene median build times over seven clean temporary rebuilds is 1.822 s. Canonically serialized source records contain 66,937 estimated tokens and the constructed node records contain 179,817, using $\lfloor \text{characters}/4 \rfloor$ per record. These latter values measure local representation size, not tokens consumed by graph construction or API billing. The manual-annotation artifact workload is measured as 97 records, 1,066 semantic items, 267,877 canonical characters, and 66,937 input token-equivalents. Together with the 179,817 output token-equivalents, the serialized build I/O is 246,754 token-equivalents. Historical annotator wall-clock time was not logged, so we do not infer a person-hour value.

4.2 Internal digital-twin pilots

Five manually captured scenes (Conference Hall, Museum, Theater, outdoor drone/street; 97 views) provide digital-twin prototype evidence (Fig. 3). Metrics are views checked, estimated context tokens, and hit@k on verified view labels. The frozen internal run sums $\lfloor \text{characters}/4 \rfloor$ over serialized view records; public-pilot records round total prompt characters divided by four. These deterministic estimates are context-size proxies, not provider billing usage. Manual annotations are *reference labels*, not independent public GT.

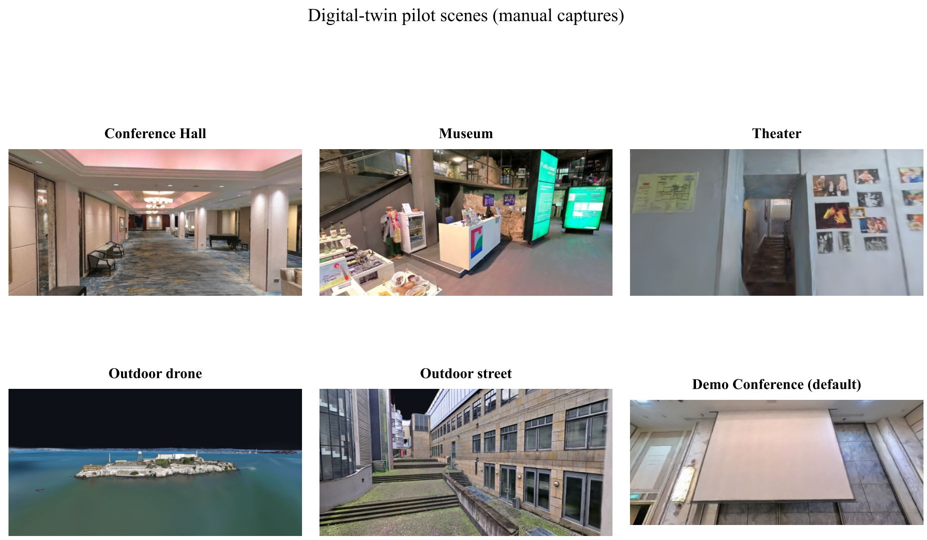

Fig. 3: Captured digital-twin pilot scenes used in the internal five-scene track.

4.3 Public-dataset pilots

We use the BBQ scene subset where possible: Replica room0–2 and office0–4; ScanNet 0011_00, 0030_00, 0046_00, 0086_00, 0222_00, 0378_00, 0389_00, 0435_00. Pilot tracks:

- Replica: 56 queries, 8 scenes, official GT object boxes (oracle semantic candidates).
- ScanNet: 48 official Nr3D/Sr3D+ queries, 8 scenes.
- External baselines: ConceptGraphs on all 8 ScanNet scenes / 48 queries and all 8 Replica scenes / 56 queries; LangSplat end-to-end on ScanNet 0011_00 / 6 queries.

We report BBQ-aligned Recall@1 and Acc@0.1 / 0.25 / 0.5. The segmentation metrics mean accuracy, mean IoU, and frequency-weighted mean IoU are **N/A** because this track has no paired model predictions. Recall@1 requires an exact target object-ID match; Acc@τ requires the predicted and GT 3D boxes to have IoU $\geq \tau$. Missing or invalid predictions remain in the eligible denominator with IoU zero.

5 Experiments

5.1 Internal five-scene graph vs. flat

Across 150 queries, graph traversal reduces average views checked from 19.4 to 4.77 (75.5%) and estimated context tokens from 3168 to 1014 (68.2%). A paired 10,000-resample query bootstrap gives 95% intervals of $[73.2, 77.7]\%$ for view savings and $[65.3, 71.0]\%$ for token savings. Cumulative estimated tokens are 475k flat vs. 152k graph (Fig. 5). On 125 GT queries with verified view IDs, stub-lexical hit@1 is 0.68 (graph) vs. 0.768 (flat) and hit@3 is 0.808 vs. 0.928 (Fig. 4). Paired graph-minus-flat intervals are $[-0.184, 0.008]$ for hit@1 and $[-0.184, -0.056]$ for hit@3. Thus hierarchy buys large context savings with a measurable hit@k tradeoff under the current stub matcher.

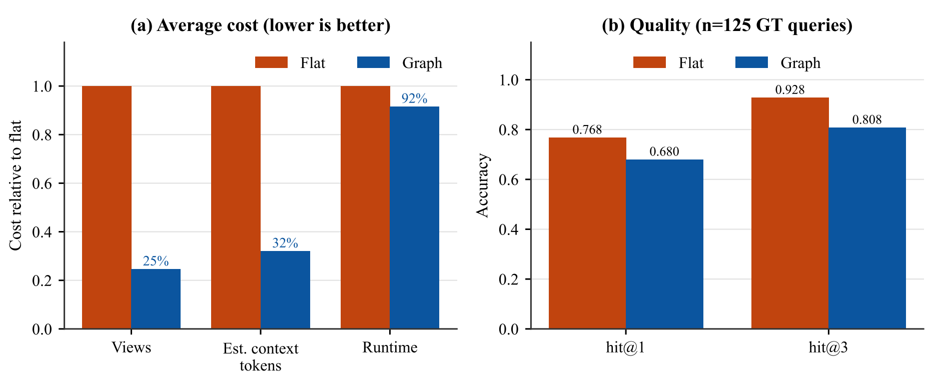

Fig. 4: Internal five-scene cost (normalized to flat) and quality (stub lexical; frozen `metrics_summary.json`).

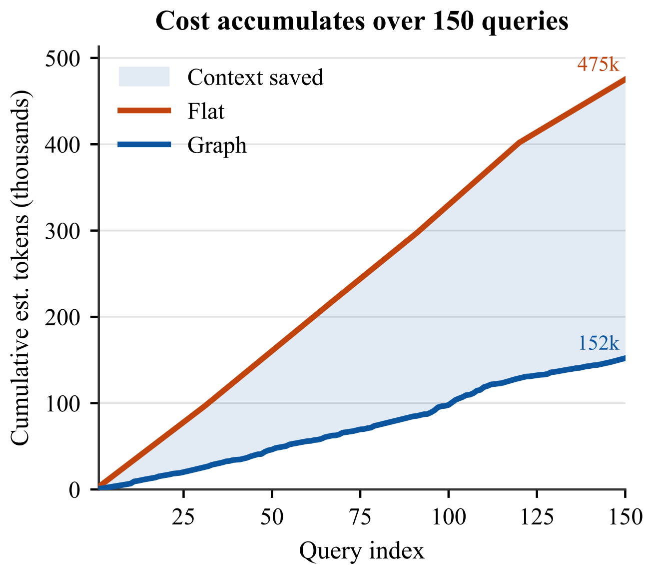

Fig. 5: Cumulative estimated context tokens over 150 internal queries (475k flat vs. 152k graph).

5.2 Where hierarchy helps

Per-query-type and per-scene analysis (Figs. 6–7) show larger savings when queries localize to one zone (e.g. negatives, simple objects) and smaller savings for multi-hop and flatter outdoor trees.

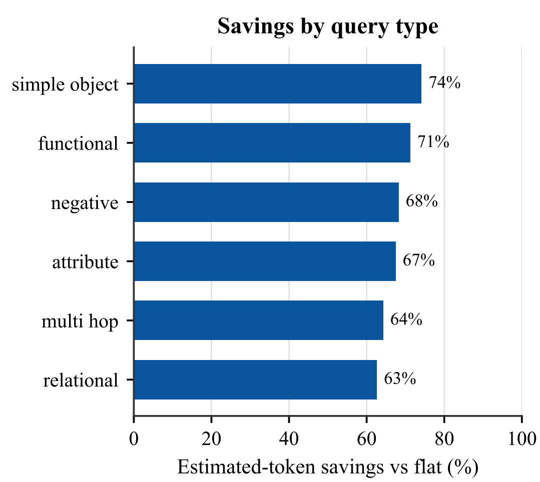

Fig. 6: Estimated context-token savings by query type on the internal five-scene track.

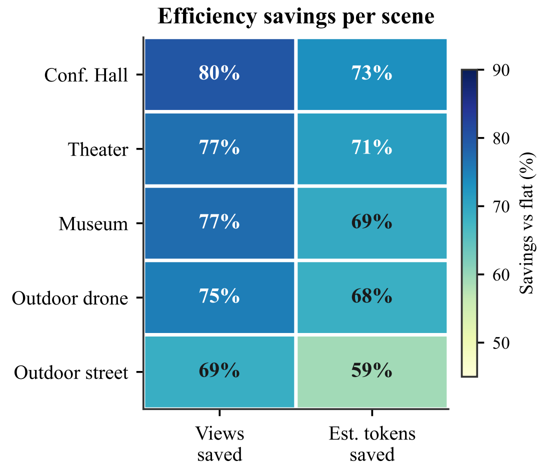

Fig. 7: Per-scene view/estimated-token savings heatmap (internal track).

5.3 Public-dataset BBQ-aligned pilots

Table 1 summarizes the frozen public-dataset runs. On Replica, graph and flat lexical reach a Recall@1 / Acc@k ceiling of 1.0 on oracle candidates—a protocol sanity check, *not* independent perception accuracy. Graph checks 12.1 objects vs. 71.9 for flat lexical. On the harder ScanNet language pilot, graph matches flat lexical quality (Recall@1 = Acc@0.25 = 0.271) while checking 11.2 vs. 49.0 objects ($\approx$ 77% fewer). Relation reranking does not help this pilot (no-relation Acc@0.25 = 0.292). Figure 8 visualizes the ScanNet quality/cost trade-off; Table 1 reports Replica oracle search cost under an Acc@k=1.0 ceiling.

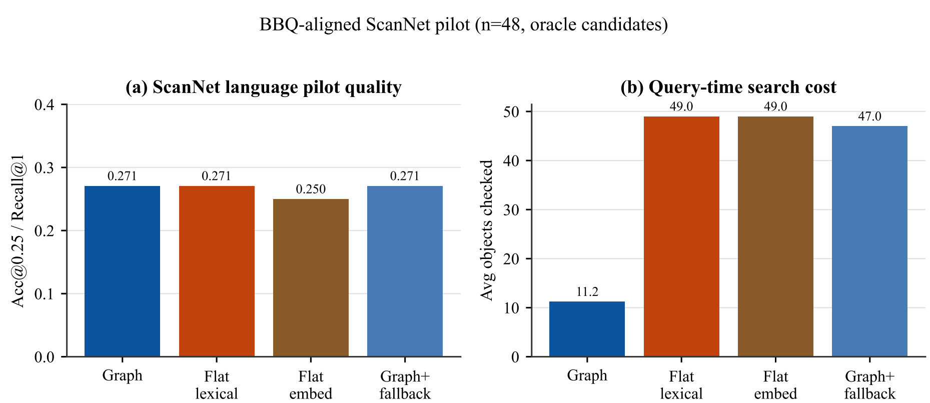

Fig. 8: ScanNet BBQ-aligned pilot (n=48 Nr3D/Sr3D+ queries, oracle candidates). Graph preserves flat-lexical Acc@0.25 while checking far fewer objects; fallback approaches flat cost.

Table 1: Public-dataset pilots (oracle semantic candidates). Efficiency = avg checked objects.

Track	Variant	Recall@1	Acc@0.25	Objects	Est. tokens
Replica (n=56)	graph	1.00	1.00	12.1	303
	flat lexical	1.00	1.00	71.9	1765
	flat embedding	0.83	0.88	71.9	1765
ScanNet (n=48)	graph	0.271	0.271	11.2	501
	graph+fallback	0.271	0.271	47.0	2027
	flat lexical	0.271	0.271	49.0	2117
	flat embedding	0.250	0.250	49.0	2117

5.4 Case studies

Conference Hall case studies (Fig. 9) show matched correctness at far lower cost for isolated targets, almost flat cost for zone-wide answers, and cheap shared misses when lexical retrieval fails for both methods. The qualitative pair prefers stage view v018 over distractor floor view v012.

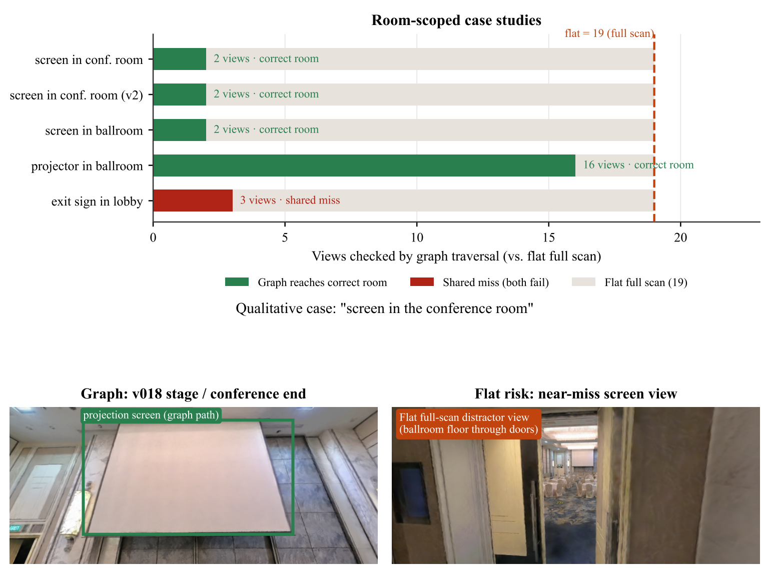

Fig. 9: Top: graph views checked versus a flat full scan. Bottom: qualitative demo views for a conference-screen query, with a view-annotation box.

5.5 External baseline executions

Table 2 reports native public-data executions. ConceptGraphs constructs predicted object maps from 39 ScanNet and 40 evenly sampled Replica RGB-D views and emits a 3D box for every positive query; Acc@0.25 is 0.0625 on both tracks. LangSplat completes RGB 3DGS optimization, SAM/CLIP preprocessing, autoencoder compression, language-field optimization, rendering, and retrieval on scene 0011_00. It selects an expected view for 4/6 queries, but emits relevance points rather than 3D boxes, so Acc@k is N/A. Token counts are total local text-encoder tokens, not API billing.

Table 2: Native external runs. Protocols and map inputs differ from the oracle-map SemanticSplat pilot; values do not define a method ranking.

Method	S	Q	Map	Quality	s/query	Tokens
CG–ScanNet	8	48	predicted	Acc@.25 = .063	.336	627
CG–Replica	8	56	predicted	Acc@.25 = .063	.274	459
LangSplat	1	6	learned field	view hit = .667	2.037	74

ConceptGraphs object-label hit / mean 3D IoU is 8/48 / 0.0696 on ScanNet and 7/48 / 0.0624 on Replica. Its unthresholded retriever returning an object for every positive query is not accuracy. The Replica maps use five evenly spaced frames per scene, not all 2,000 trajectory frames. LangSplat uses 3,000 RGB and language iterations with SAM-B / CLIP-B on a 6 GB GPU, rather than the paper's 30,000-iteration / 24 GB setting. BBQ official code is **not run**; its tables remain related-work evidence only [9].

6 Limitations

- Manual five-scene annotations are reference labels, not independent public GT.
- Internal stub-lexical hit@k currently favors flat over graph (hit@1 0.68 vs. 0.768); public pilots match quality while cutting objects checked.
- Public pilots use oracle semantic candidates; Replica Acc@k=1.0 is a ceiling sanity result.
- mAcc/mIoU/fmIoU are unavailable without predicted segmentation arrays.
- Relation reranking currently underperforms its ablation on ScanNet pilot.
- No same-protocol BBQ official-code comparison; no superiority claim.
- External executions are not matched comparisons: ConceptGraphs predicts objects from sampled RGB-D frames, LangSplat learns a one-scene field, and SemanticSplat public pilots search oracle semantic candidates.
- Stub / deterministic lexical matching is not live provider-backed VLM evaluation.

7 Reproducibility

All reported values are generated from frozen machine-readable aggregate and per-query results. Run configurations record scene and query identifiers, variant flags, model provenance, and one-time construction cost; figure scripts consume those same outputs. An automated consistency gate verifies the paper values against the frozen JSON. Paired percentile intervals use 10,000 query resamples with seed 20,260,715. No external URL or supplementary result is required to interpret the claims in this review manuscript.

8 Conclusion

SemanticSplat shows that hierarchical graph pruning reduces query-time context for 3DGS digital twins under a controlled same-input protocol. On the internal five-scene track the savings are large (75.5% views / 68.2% estimated context tokens) with a stub-lexical hit@k tradeoff; on BBQ-aligned ScanNet/Replica pilots, Acc@k matches flat lexical while checked objects drop sharply. We do not claim BBQ superiority. Future work: stronger relation-aware search, predicted segmentation track, and fair same-protocol BBQ execution if permitted.

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